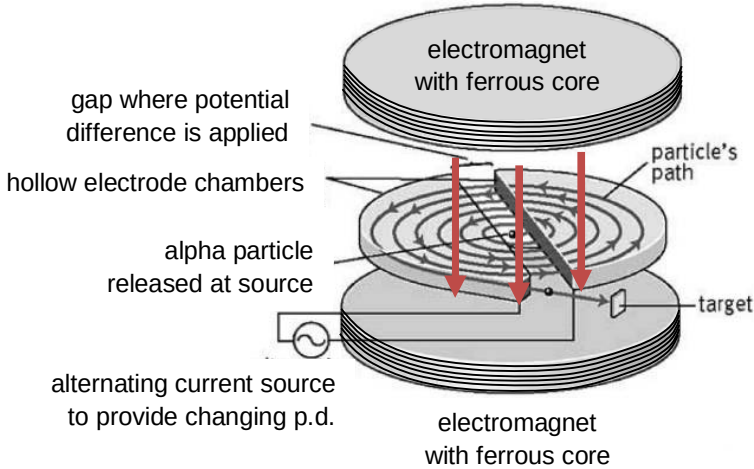


4	(a)		<u>region of space where a magnetic pole (or magnet), moving charged particle or current-carrying conductor will experience a magnetic force</u>	
	(b)		 The diagram illustrates the internal structure of a cyclotron. It features two semi-circular hollow electrode chambers, known as dees, which are positioned face-to-face. These chambers are held together by two large electromagnets with ferrous cores. The gap between the dees is where a potential difference is applied. An alpha particle, released from a source at the center, follows a spiral path (labeled 'particle's path') as it is accelerated back and forth between the dees by the alternating electric field. The dees are connected to an alternating current source to provide a changing potential difference. The particle's path terminates at a target located at the edge of one of the dees.	

	(c)	(i)	Magnetic force provides for centripetal force for particles to move in a circular path: $\frac{mv^2}{r} = qvB$ $r = \frac{mv}{qB}$ Mass of alpha particle = 4u Charge of alpha particle = +2e $r = \frac{4uv}{2eB} = \frac{2uv}{eB}$ Correct substitution of both mass and charge in terms of u and e	
		(ii)	Only dependent variable is velocity as B, u and e are constant. Each time ions pass between, they are <u>accelerated by the electric field</u> due to potential difference between the dees. OR <u>gains kinetic energy</u> from potential difference (from electric potential energy) The <u>velocity increases, therefore radius increases.</u>	
	(d)		Magnetic flux density at centre of solenoid is given by $B = \mu_0 n I$ $B = (4\pi \times 10^{-7}) \left(\frac{3100}{0.39} \right) (200)$ $B = 1.998 \text{ T}$ $= 2.0 \text{ T (shown)}$	

	(e)	Magnetic force provides for centripetal force for particles to move in a circular path: $mv\omega = qvB$ $\omega = \frac{qB}{m}$ $T = \frac{2\pi m}{qB}$ $T = \frac{2\pi(4)(1.66 \times 10^{-27})}{2(1.6 \times 10^{-19})(2.0)}$ $T = 6.519 \times 10^{-8} \text{ s}$ Every complete circle/cycle, the particle is accelerated twice Total number of times accelerated $= 0.52 \times 10^{-3} / 6.519 \times 10^{-8} \times 2$ $= 15954 \text{ times}$ Energy gained per acceleration across dees $= \frac{(1.6 \times 10^6)(1.6 \times 10^{-19})}{15954}$ $= 1.6047 \times 10^{-16} \text{ J}$ Energy gained = charge x potential difference p.d. $= \frac{1.6047 \times 10^{-16}}{(2)(1.6 \times 10^{-19})}$ $= 501.5 \text{ V}$ $= 500 \text{ V (2 sf)}$	
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5	(c)	(i)	Work function is defined as the minimum energy required to eject an	
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